# Denoising Lunar Surface Images in order to find proper landing zones using Variational Auto-Encoders Transfer Learning and Riemannian Geometry.

*Vasileios Feidantsis ph5320*

# Project overview:

- Goal: Develop a model that reconstructs lunar images and detect smooth regions for safe landings.

- Motivation: Importance of precise landing sites for lunar missions.

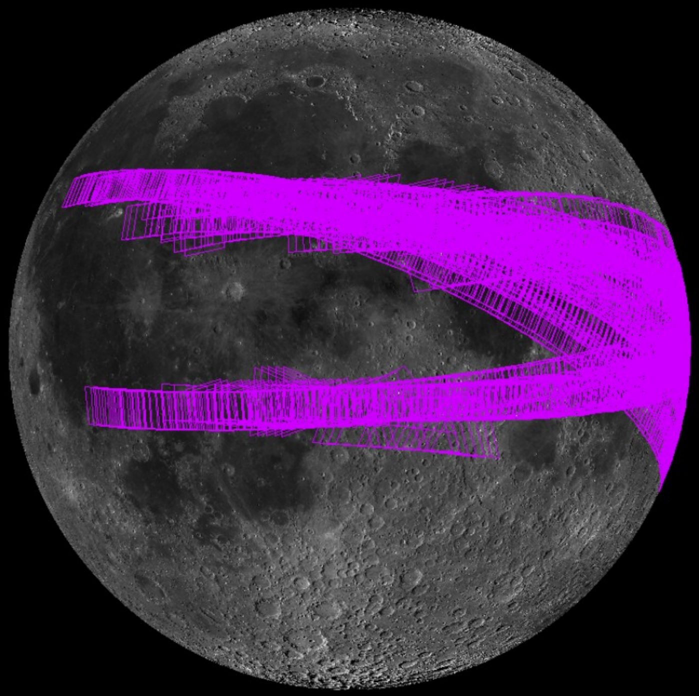
- Approach: Use of VAE with transfer learning and smoothness analysis with Riemannian Hypersphere Geometry in the latent space.

# Database:

- Image database taken from im-ldi.com

- It contains images from several missions that were used to map out the surface of the moon

- In our case the data were extracted from the Apollo Metric images.



# Database: Filters chosen

- The entire database consisted of 5000 (due to the sites limitations).

- However in order to get images with adequate lighting I filtered the dataset to obtain only the ones with a Local Sun Location (0-90°) of 33°-68° .

- This choice made us end up with 2511 images

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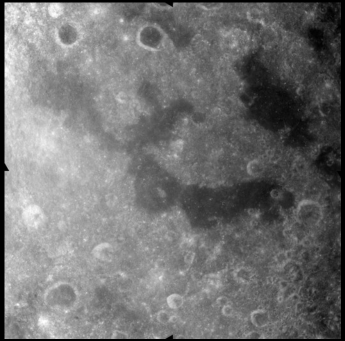

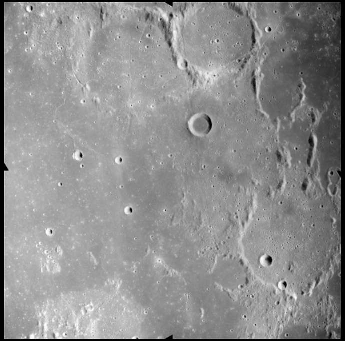

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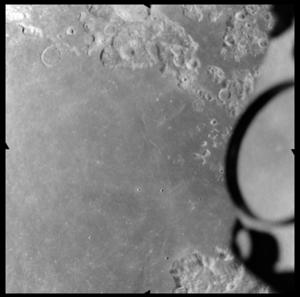
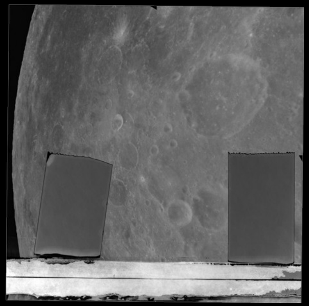
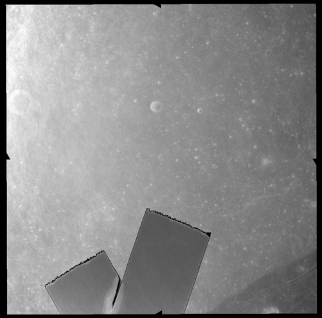
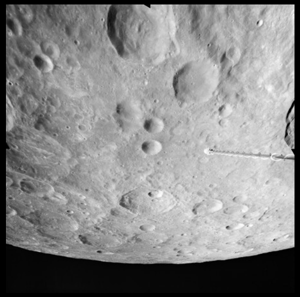






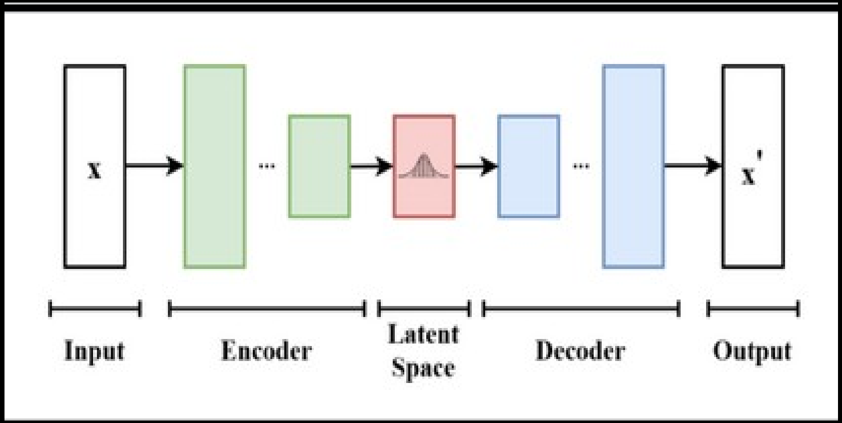


# Methodology: Model Selection

VAEs are generative models that combine an encoder and a decoder.

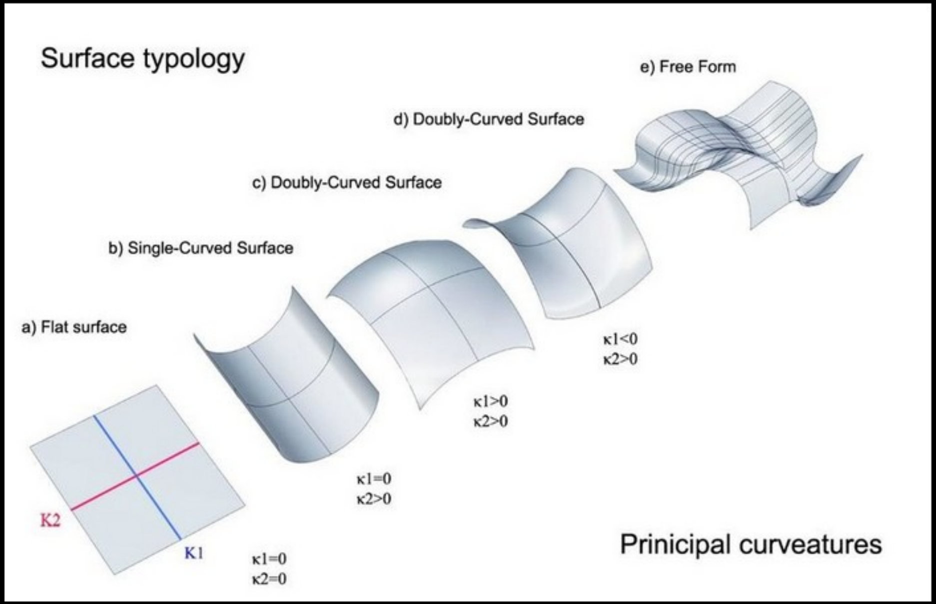
Key Components:

- Encoder: Maps input to latent space.
- Decoder: Reconstructs image from latent space



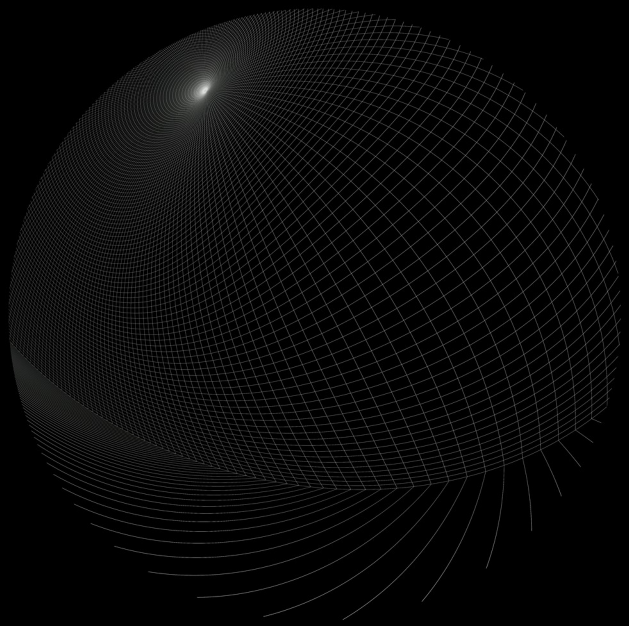
# Methodology: The problem with Standard VAEs:

- A standard VAE maps images to a Euclidean latent space (a flat, grid-like space).
- However, complex data like lunar terrain naturally exists on a curved manifold.
- Forcing this data into a flat space can distort relationships between images and lead to a less efficient representation.



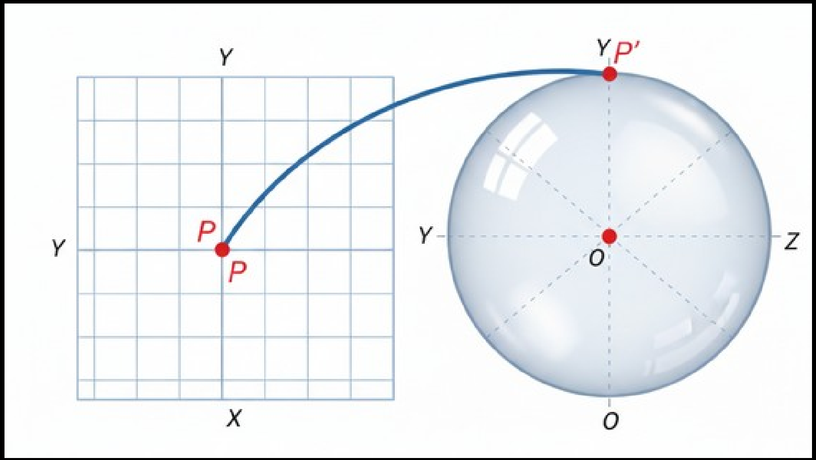
# Methodology: A Riemannian Latent Space

- We used Riemannian geometry to design a latent space that is not flat.
- A hypersphere serves as our curved latent space, which better preserves the natural relationships in the data.
- This allows the model to learn a more organized and coherent representation of the lunar surface.



# Methodology: The Hyperspherical VAE:

- The *HypersphereSampling* layer is the key component that implements this.
- The encoder projects the data onto the surface of a unit hypersphere.
- This forces all learned latent representations to be confined to this curved manifold.



# Methodology: The Mathematical Insight

- We replaced the standard KL divergence loss with a custom *hypersphere_kl_divergence* function.
- This new loss function encourages the learned distribution to be uniform on the hypersphere, ensuring the entire latent space is used effectively.

$$L_{KL} = \frac{1}{N} \sum_{i=1}^{n} \left( \kappa_i - \log(\kappa_i) \right)$$

# Methodology: Model Architecture

For our intends and purposes two models were chosen, namely a Hyperspherical VAE, and a Complex VAE.

Hyperspherical VAE:

- Input 512x512x1
- Layers: Conv2D, BatchNorm, Dense, Conv2DTranspose
- Hyperspherical Latent Space

Complex (U-Net like) VAE:

- Adds skip connections (64x64,32x32,16x16,8x8)
- Transfer Learning: Fine-Tuning the model with weights from the Hyperspherical VAE..
- Reconstruction Enhancement: We use VGG16 (pre-trained on ImageNet) for perceptual loss and for Feature Extraction

# Methodology: Training Process with Transfer Learning

Data pre-processing

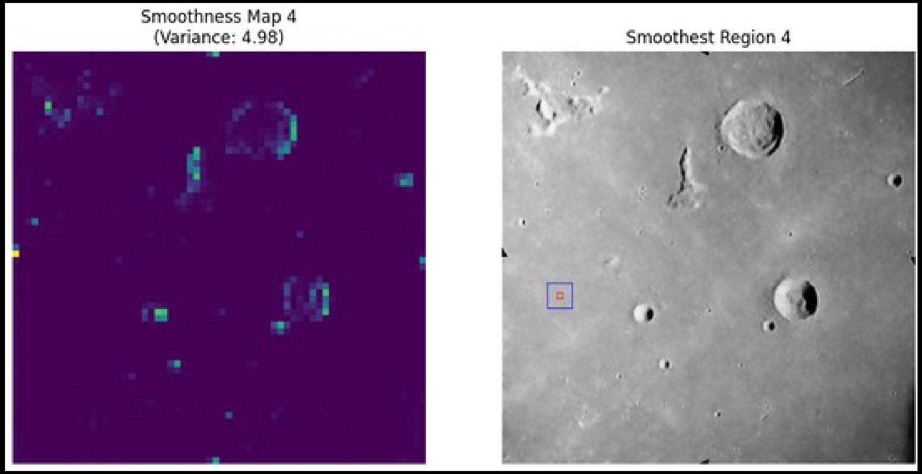
- Resize to 512x512
- Normalise to [0,1]
- Add Noise factor (0.1)
- Train-Test split:80%-20%

Training Details:

- Optimizer Adam: lr=0.0005
- Loss: MSE+Perceptual Loss+ KL Divergence (respective to each VAE)
- Callbacks: EarlyStopping, ReduceLROnPlateau
- Transfer Learning: Fine-Tune complex VAE with Hyperspherical VAE weights
- Epochs: 100 (on each)

# Methodology: Smoothness calculation

- In order to determine the final landing zones based on the reconstructed images, we are going to use a variance map based on the variance of pixel values.
- Based on how the lunar surface is lit, the pixel values reflect whether the terrain is uniform or if it exhibits inconsistencies.
- Also after landing we want to make sure that a rover will be able to navigate through the LZ to reach its target area.
- That's why we are going to use 2 windows. An 8x8 pixel one to determine the smoothness of the LZ, and a 32x32 one to ensure that there are possible paths for the rover to navigate.



# Results: Training Results Reconstruction Loss (0.3 noise)

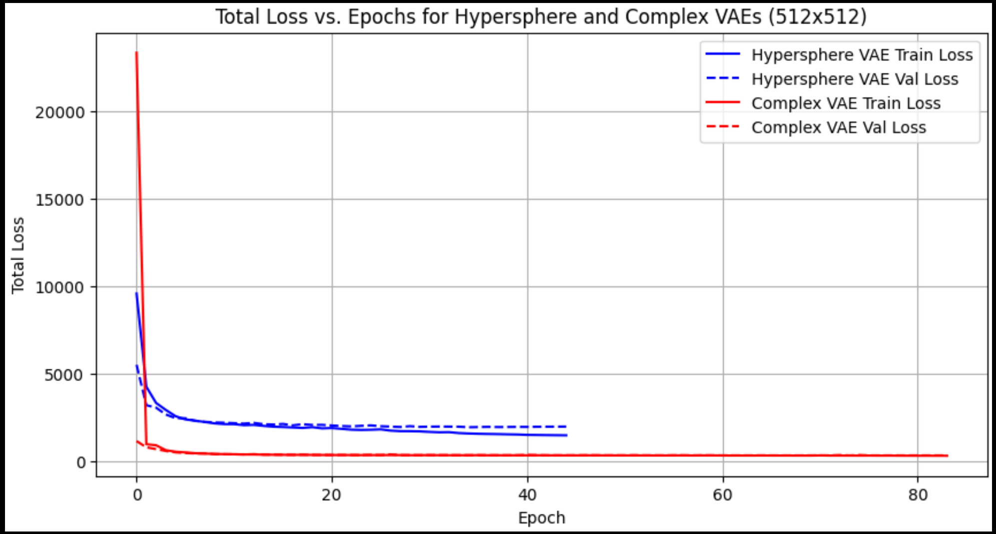
## Loss Trends:

- **Hyperspherical VAE Reconstruction Loss:**

  *(1464 on the training set & 1960 on the validation set)*

- **Complex VAE Reconstruction Loss:**

  *(317 on the training set & 338 on the validation set)*



# Results: Training Results PNSR (0.3 noise)

- ***Original Image vs Noisy PNSR:***

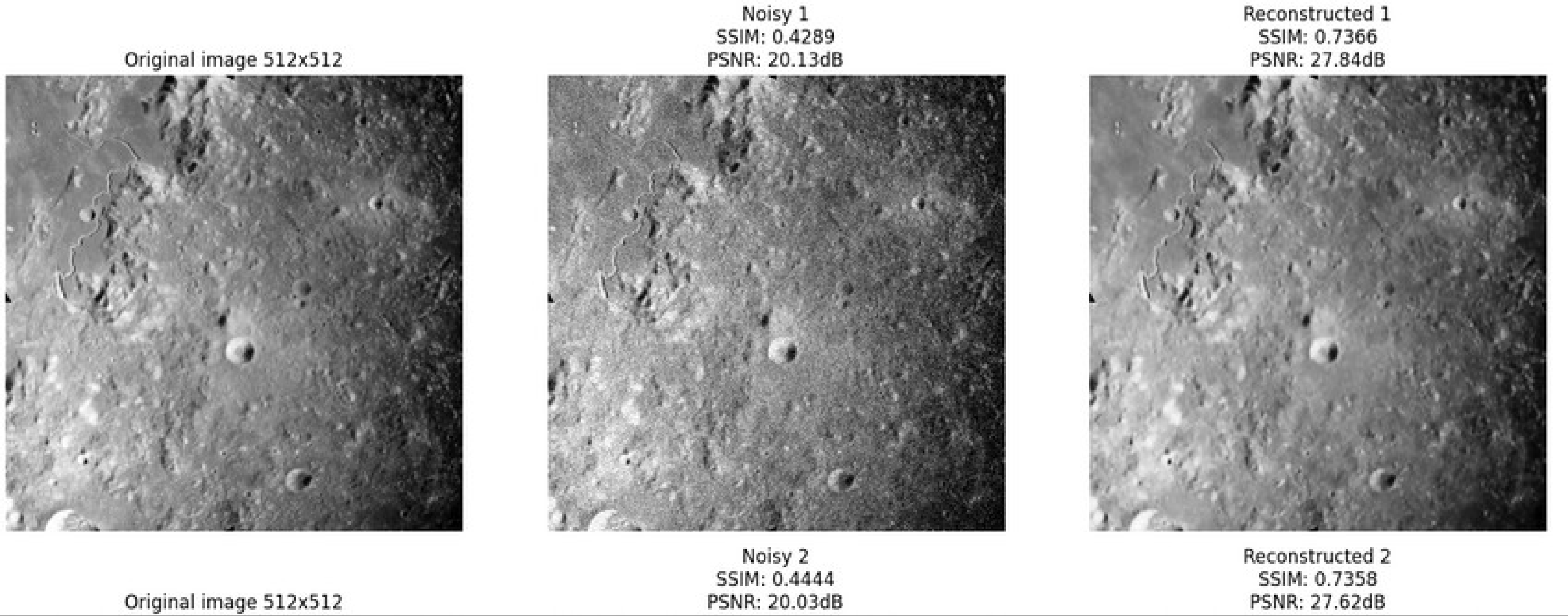
  *(28.11dB)*

- ***Original Image vs Reconstructed PNSR:***

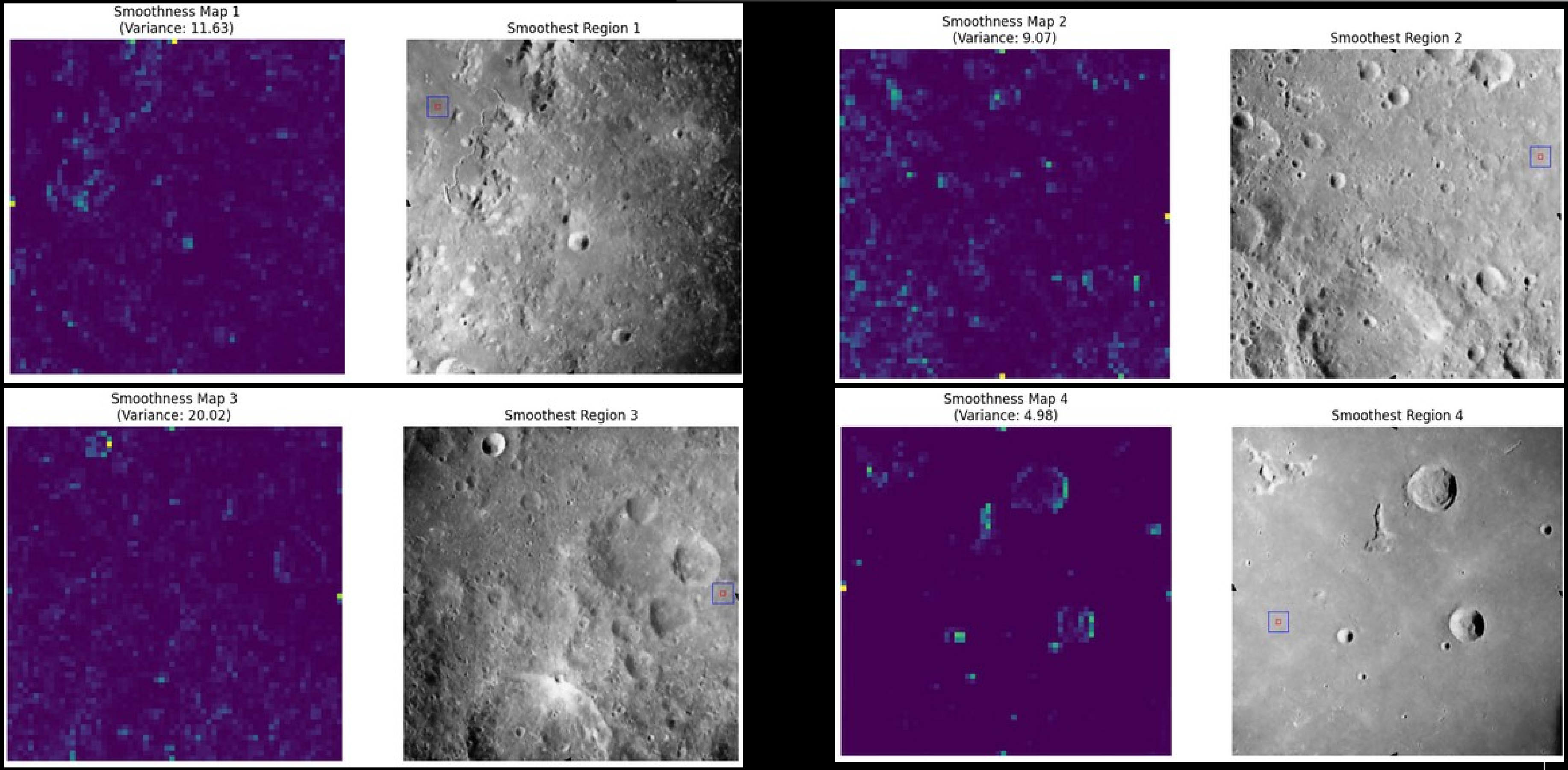
  *(20.05dB)*

- *85% noise elimination*

# Results: Image Results regarding Reconstruction



# Results: Image Results regarding LZ selection



# Limitations:

The main limitations to getting better results were:

- Hardware limitations (16GB VRAM GPU on a local setup despite optimizations)
- Small train test set (500 images)
- Rescaling from 1012x1012 to 512x512
- The assumption that uniform resizing preserves features

# Future Considerations:

- Train on a larger dataset.
- Train on images with a higher resolution.
- Real-Time processing.

# Conclusion:

To summarise,

- A VAE with transfer learning VGG16 and Riemannian Geometry reconstructs and identifies smooth lunar regions with noteworthy success.
- This could support safer, and efficient lunar missions.

Thank you for your attention

Vasileios Feidantsis